

No weights, logits, or training data required

Target LLM

Binary responses only

$$\mathbf{x} = (x_1, \dots, x_n)$$

PSN-IRT Item Bank

Pre-calibrated 4PL

$$(a_i, b_i, c_i, d_i)$$

ℓ_z^* Person-Fit Analysis

$\hat{\theta}_{\text{MLE}}$ ability estimation

$$\Delta \ell_z^* = \ell_{z, \text{test}}^* - \bar{\ell}_{z, \text{clean}}^*$$

Matched-difference design

Detection

Cohen's d effect size

Bootstrap 95% CI

Diagnosis

θ -shift analysis

Seen/unseen decomposition